RRC-Raptor-V4H MSIOF

- User’s Manual: Software

V0.0.1

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i. VERSION LIST

Description	Version	Date	Author
RTX_MSIOF	V0.0.1	2024/07/03	Jerry

ii. HASHTAG

#V4H
#MSIOF
#SPI
#BSP

1. Overview

Target：介紹如何修改 Device Tree，以及如何使用 spidev-test 工具讀寫 spi 裝置..

2. Directory Configuration

The directory configuration is shown below.

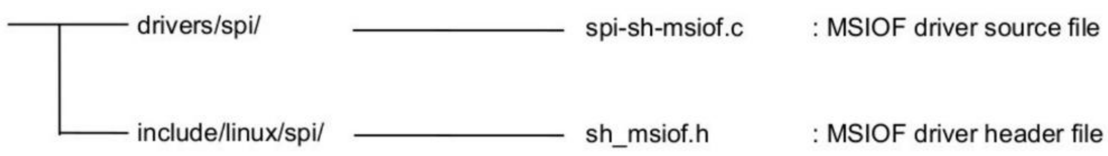


Figure 1. Directory Configuration

3. Pin Function setting

The editing contents are shown in Figure 2 (Example for r8a779g0-raptor.dts)

```
&pfc {
    msiof1_pins: msiof1 {
        groups = "msiof1_clk", "msiof1_rxd", "msiof1_txd", "msiof1_sync", "msiof1_ss1", "msiof1_ss2";
        function = "msiof1";
    };

    msiof2_pins: msiof2 {
        groups = "msiof2_clk", "msiof2_rxd", "msiof2_txd", "msiof2_sync", "msiof2_ss1", "msiof2_ss2";
        function = "msiof2";
    };
};
```

Figure 2. Pin Function setting

4. Channel Node setting

Enable MSIOF1 status property for R-Car V4H Raptor.

```
&msiof1 {
    pinctrl-0 = <&msiof1_pins>;
    pinctrl-names = "default";
    num-cs = <3>;

    status = "okay";

    slavedev1@0 {
        compatible = "maker,slavedev";
        reg = <0>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };

    slavedev1@1 {
        compatible = "maker,slavedev";
        reg = <1>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };

    slavedev1@2 {
        compatible = "maker,slavedev";
        reg = <2>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };
};
```

Figure 3 Channel Node setting

Enable MSIOF2 status property for R-Car V4H Raptor.

```
&msiof2 {
    pinctrl-0 = <&msiof2_pins>;
    pinctrl-names = "default";
    num-cs = <3>;

    status = "okay";

    slavedev2@0 {
        compatible = "maker,slavedev";
        reg = <0>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };

    slavedev2@1 {
        compatible = "maker,slavedev";
        reg = <1>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };

    slavedev2@2 {
        compatible = "maker,slavedev";
        reg = <2>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };
};
```

Figure 4 Channel Node setting

5. Kernel configuration

To enable the function of this module, make the following setting with Kernel Configuration. The changes affect after re-building the kernel.

```
Device Drivers    --->
        [*] SPI support    -->
                --- SPI support
                 <*>   SuperH MSIOF SPI Controller
                 [*]       Transfer Synchronization Debug support for MSIOF
                 (1)          Master of sleep latency (msec time)
                       ...
                       *** SPI Protocol Masters ***
                 <*>   User mode SPI device driver support
                       ...
                 [ ]   SPI slave protocol handlers *1
```

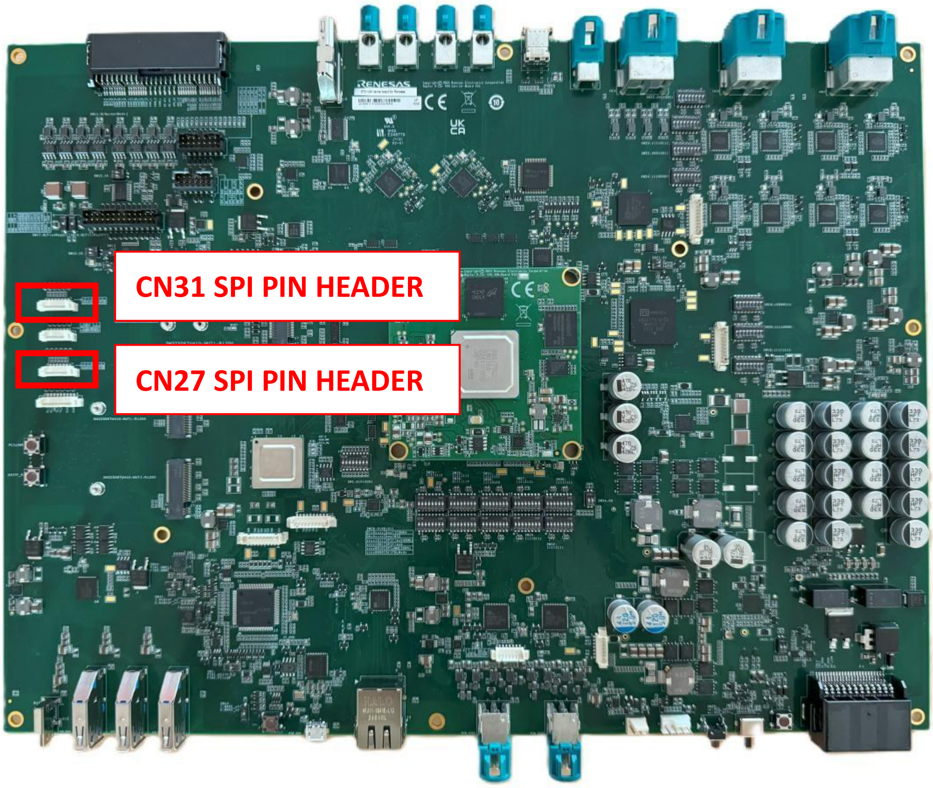
Figure 5 Kernel configuration

Notes:

*1: Please set if you want to enable SPI Slave mode.

6. Connect

提供 MSIOF interface 讓 User 可以自己加入 SPI device



CN31 : SPI PIN HEADER (**spidev0**)

PIN	SYMBOL
1	MSIOF1_SS2/MD18_SW (**CS2**)
2	MSIOF1_SS1/MD16_SW (**CS1**)
3	MSIOF1_SCK/MD14_SW (**SLK**)
4	MSIOF1_SYNC/MD32_SW (**CS0**)
5	MSIOF1_RXD/MD11_SW (**DO**)
6	MSIOF1_TXD/MD13_SW (**DI**)
7	GND

CN27 : SPI PIN HEADER (**spidev1**)

PIN	**SYMBOL**
1	MSIOF2_SS2/MD18_SW (**CS2**)
2	MSIOF2_SS1/MD16_SW (**CS1**)
3	MSIOF2_SCK/MD14_SW (**SLK**)
4	MSIOF2_SYNC/MD32_SW (**CS0**)
5	MSIOF2_RXD/MD11_SW (**DO**)
6	MSIOF2_TXD/MD13_SW (**DI**)
7	GND

7. SPI Tool

Use spidev-test can read / write spi device.

I. Here is an example of Raptot connecting to SPI Flash through msiof1 (CN 31).

```
root@v4x:~# spidev_test -HOv -D /dev/spidev0.0 -p '\x90\xff\xff\x00\x00\x00'
spi mode: 0x3
bits per word: 8
max speed: 500000 Hz (500 kHz)
TX | 90 FF FF 00 00 00 __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __  |......|
RX | 00 00 00 00 EF 15 __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __  |......|
```

Figure 6 example of spidev-test method

II. 如果要用 msiof2 (CN 27), 則將 command 改為

spidev_test -HOv -D /dev/spidev1.0 -p '\x90\xff\xff\x00\x00\x00'

8. APPENDIX